2019 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.202 Human-Computer Interaction

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a One of the four key Interaction Types is 'Manipulation' whereby users interact with objects in a virtual or physical space by manipulating them. Describe each of the other three key Interaction Types and for each one provide an example of a benefit that comes from using it.

[9 marks]

1.b Ben Shneiderman coined the term 'Direct Manipulation'. Briefly describe what you understand by this term and its core principles. Include concrete examples when describing the core principles.

[7 marks]

1.c Briefly explain one of the ways in which Direct manipulation can provide support for novice users.

[1 marks]

1.d Briefly describe one of the disadvantages of using Direct manipulation.

[1 marks]

1.e You work for a company that is considering the development of a sensor-equipped glove which when worn by a user enables him/her to point at a Point of Interest, such as an historic building, in order to receive information about it via their smartphone. The primary target audience for the glove is tourists.

While technically feasible, the development of a fully working prototype glove would clearly be expensive. Before committing to this full expense the company decides to carry out a Wizard of Oz study in order to gauge user reaction to such a glove.

Describe what you understand by the term 'Wizard of Oz study' and explain how you would carry out such a Wizard of Oz study in the city of Lancaster.

[6 marks]

1.f Effectively the sensor-equipped glove allows the user's orientation to be used for the purpose of information retrieval. Orientation is one of the 5 basic dimensions considered under the concept of proxemics discussed in lectures. Which of the other 5 basic dimensions could also be considered as supporting information retrieval in this scenario? Briefly explain your answer.

[1 mark]

Total 25 marks

Question 2

In your role as an Interaction Designer you have been commissioned by a company to produce a prototype smart 'pill dispenser' (and associated mobile app) for use in the domestic home setting.

2.a You have been asked to follow a UCD approach. Briefly describe what you understand by this term and its key strengths and weaknesses. How does it compare with a System Centered Design approach?

[3 marks]

2.b The company has allowed you three weeks to carry out a needs-finding study. It has recruited five suitable households that have agreed to participate for the entire study. Describe AND JUSTIFY two methods you consider appropriate for capturing data in these domestic settings and one which you consider inappropriate. Also, briefly explain why running the study for three weeks is more appropriate than running the study for a single day.

[7 marks]

2.c At the end of the three weeks you are invited back to the company in order to present some of your findings. You decide to make use of Personas and Scenarios in your presentation. Explain what you understand by these two terms and how they could be effective in helping you communicate your findings?

[2 marks]

2.d At the end of your presentation the company agree that there is value in continuing with the design and commission you to carry out the next phase which focusses on the use of paper prototyping in order to produce a number of candidate designs (of both the medical dispenser and the accompanying app). Explain what you understand by the term paper prototyping and describe one of its key advantages.

[2 marks]

2.e Describe how you plan to have the paper prototypes evaluated.

[2 marks]

2.f One of the key functions that need to be supported by the mobile app is a 'set time between dispensing' function. Following the paper prototyping evaluation you have two candidate UI designs for supporting this function. You decide to produce a vertical prototype for each of the candidate UI designs. What is a vertical prototype and why is it an appropriate approach to follow at this stage.

[2 marks]

Question 2 continues on next page…

Question 2 continued.

2.g You carry out a usability study of the two vertical prototypes, P1 and P2. As part of the study users are asked to carry out a number of tasks using the 'set the time between dispensing' function. These tasks include changing the time to a range of different specified values. The results of the study are shown in the table below. Based on these results, which prototype would you recommend taking forward. Explain your reasoning.

Table 1. Mean time on task and error rate for prototypes P1 and P2

	Time on Task (s)	Error Rate (%)
P1	2.3 (+-1.5)	2.8 (+-0.8)
P2	3.1 (+-1.6)	5.2 (+-1.3)
t-test	p = .0713	p=.0278

[3 marks]

2.h Once a fully working prototype has been developed you suggest that a longitudinal evaluation be carried out in the five households. What is the minimum period of time that you would recommend for such an evaluation, and why? Explain the factors that need to be taken into account. What additional data capture method could be employed during this evaluation in order to help understand the use of the prototype in a realistic setting?

[4 marks]

Total 25 marks

Question 3

3.a WIMP is the short form for a user interface paradigm. Which four concepts does the acronym refer to? For each of the concepts, describe in one sentence the user interface problem it solves. (You only need to state the problem and are not expected to explain how it is tackled.)

[5 marks]

3.b Graphical user interfaces are based on a view hierarchy. To illustrate the concept, sketch a suitable part of the user interface you (co-)designed in your coursework and draw the corresponding view hierarchy next to it. The hierarchy must have at least three levels. Describe briefly what view hierarchies are used for in the user interface.

[6 marks]

3.c Surround.com aim to develop a user interface that extends the desktop metaphor for use in Head-Mounted Displays (HMDs). The desktop height would be fixed, but horizontally it will completely surround the user, so that they can place icons and windows all around them. The viewport (i.e. which portion of the desktop they can see at any point in time) will be controlled by where users turn their head relative to their environment. As users may not always want to interact through an HMD, Surround.com also want to support use of the virtual desktop on a tablet computer. In this case, the viewport will be controlled by touch-and-drag action on the tablet's surface. As in the HMD, when users continue to navigate in one direction (e.g. left), they eventually get back to where they started.

Surround.com ask you to provide a detailed design for the proposed viewport control techniques, for both the HMD and the tablet interface. For each one, describe the property you would sense as input, the property you control, and the control-display relationship.

[8 marks]

3.d Discuss how the two viewport control techniques differ from the perspective of the user. Do you expect one to be faster than the other? Will they differ in accuracy? Would you think that one is easier to learn and use than the other? Explain your reasoning.

[6 marks]

Total 25 marks

Question 4

4.a Provide SHORT answers to the following questions:

i. In human cognition, what is a unit task?

ii. What is the approximate duration of a unit task?

iii. What is considered the time limit for perception of cause-effect relationships?

iv. Errors can be classified into two major types. Which are these?

v. “Preview results” is an error prevention guideline for which type of error?

vi. For which type of error is “Include helpful constraints” a prevention guideline?

vii. Identify two principally different mechanisms by which people can retrieve information from memory.

[7 marks]

4.b A general design principle is to reduce the demands an interface places on the user's short-term memory.

i. Explain concisely, in one sentence each, what short-term memory is and what limitations short-term memory has.

ii. Identify a human strategy for coping with the limitations of our working memory. Explain in concrete terms how a user interface can support this strategy.

[5 marks]

4.c Based on Psychology research published by George Miller in the 1950s, many designers have argued for a “Miller’s Law” principle of limiting the number of options shown in menus to “seven plus minus two”. Provide a critical discussion of the argument.

[5 marks]

Question 4 continues on next page…

Question 4 continued.

4.d Your team are developing a web application and have come to show you four possible designs for the registration form, shown below. The form has four text fields, and each of the designs suggests a different positioning of the associated labels, including one version where the label is shown in the field, to be typed over. Discuss for each why you would or would not choose it, with reference to relevant aspects of human cognition or perception.

[8 marks]

Total 25 marks

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